

Foundations

Data Can Deceive · Causal Roadmaps with (and without) AI

Module C

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Association versus Intervention

Association – *seeing*

- Patterns already in the data
- Treatment and outcome move together
- Purely descriptive: $P(Y \mid A)$

Intervention – *doing*

- What happens if *we act*
- Outcome under a hypothetical change
- Requires assumptions: $P(Y \mid do(A))$

An observed association can come from:

- a genuine causal effect,
- confounding (a common cause),
- selection, measurement, or other bias.

The task: move from what we see to what would happen if we *acted*, with the assumptions made explicit.

Pearl's Ladder of Causation

Three levels of causal reasoning [3]:

1. **Association (Seeing)**

What does observing A tell me about Y ? $P(Y | A)$

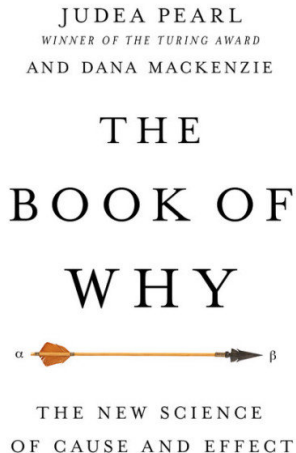
2. **Intervention (Doing)**

What happens to Y if I *do* A ? $P(Y | do(A))$

3. **Counterfactuals (Imagining)**

What *would have happened* had A been different?

Key insight: each rung needs information the rung below cannot supply. No amount of observational data alone answers an interventional question.



The Counterfactual Contrast

A causal effect compares outcomes under different interventions:

- $Y^{(1)}$: the outcome if a person *were* treated, $A = 1$
- $Y^{(0)}$: the outcome if that same person were *not*, $A = 0$

An average causal effect exists when $E[Y^{(1)}] \neq E[Y^{(0)}]$.

Formalized in Hernán & Robins' [2] *Causal Inference: What If*.

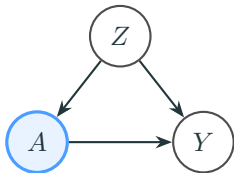
The fundamental problem of causal inference

For any one person we only ever see *one* of $Y^{(1)}, Y^{(0)}$. The other is missing, always. Causal inference is a missing-data problem we solve with design and assumptions, not with a bigger dataset.

Confounder: A Common Cause

Generic structure

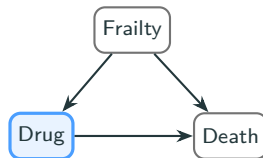
fork: $Z \rightarrow A$, $Z \rightarrow Y$



Adjust for Z to close the backdoor.

Example: frailty

sicker patients get the aggressive drug

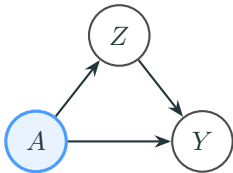


Frailty drives both treatment *and* death, so the crude contrast makes the drug look harmful.

Mediator: On the Causal Path

Generic structure

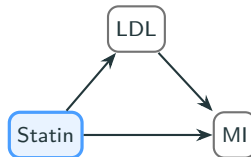
chain: $A \rightarrow Z \rightarrow Y$



Do not adjust (for a total effect).

Example: statin and LDL

the drug acts *through* lowering LDL

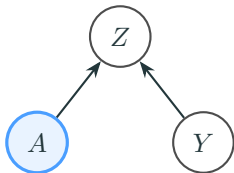


Adjust for LDL and the statin looks inert – you have removed the pathway it acts through.

Collider: A Common Effect

Generic structure

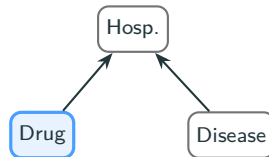
collision: $A \rightarrow Z \leftarrow Y$



Do not adjust – it opens a fake path.

Example: hospitalization

drug and disease both cause admission



Restrict to the hospitalized and drug and disease look linked.

Why “Adjust for Everything” Fails

A single reflex applied to all three does two kinds of damage:

Adjust a mediator

You block part of the effect you are trying to measure. Condition on LDL and the statin looks inert – you have thrown away the mechanism.

Adjust a collider

You *open* a path that was closed. Restrict to hospitalized patients and drug and disease look linked when in the population they are not.

The lesson: the covariate list is a set of *claims about structure*, not a bigger-is-safer knob. What to adjust for is decided by the causal question and the diagram, before the data.

Which is exactly what Segment D’s roadmap makes systematic.

Design First: The Target Trial

Specify the randomized trial you *wish* you could run [1], then emulate it with the data you have:

- eligibility, treatment strategies, assignment,
- follow-up, outcome, and the causal contrast.

Time zero: align eligibility, treatment, and follow-up at one instant

Every patient's clock starts when they meet eligibility and treatment is assigned. Getting this wrong is the classic source of **immortal-time bias** (survival smuggled into the treated group) and **prevalent-user bias** (survivors of early treatment overrepresented).

Why it matters here: a clean time zero is a *design* fix for problems no downstream model can undo.

The Pneumonia-Vaccine Paradox

Observational studies of the pneumococcal vaccine in older adults have reported that **vaccinated patients had *higher pneumonia risk*** than the unvaccinated – the vaccine looks useless, even harmful.

But look at the timing:

- In a large primary-care cohort, the vaccinated already had higher pneumonia rates *in the year before they were vaccinated*.
- An “effect” that precedes the exposure cannot be caused by it.
- The gap is a marker of *who* gets vaccinated, not of harm.

Resolution: confounding by indication

The vaccine is targeted at frailer, higher-risk patients. Baseline frailty is a **confounder** – it raises both the chance of vaccination *and* the risk of pneumonia. When the trial is actually run, the vaccine protects (CAPITA: 46% against vaccine-type pneumonia) [4].

What Would Let Us Claim Causality?

The causal question is a contrast of **counterfactuals**:

$$\Psi = E[Y^{(1)}] - E[Y^{(0)}]$$

but each patient shows us only the outcome under the exposure they actually had. No sample size fills in the missing half.

Identification: the bridge from data to counterfactuals

Three assumptions let an observed-data quantity stand in for the counterfactual contrast: **exchangeability**, **positivity**, and **consistency**.

Why it matters here: these are *design* claims, argued before any model is fit — the roadmap (Module D) is where each one is declared and defended.

Exchangeability — No Unmeasured Confounding

Within strata of the measured covariates W , exposure is as good as random:

$$Y^{(a)} \perp\!\!\!\perp A \mid W \quad \text{for } a \in \{0, 1\}$$

- In the paradox: baseline frailty drives both vaccination and pneumonia risk. Exchangeability holds only if W captures that difference.
- The DAG carries the argument: the adjustment set W must block every backdoor path.

Not checkable from the data

No statistical test can confirm exchangeability. It rests on the causal model, and its failure is probed only by sensitivity analysis.

Positivity — Everyone Could Get Either Exposure

Every kind of patient has a real chance of each exposure level:

$$0 < P(A = 1 \mid W = w) < 1 \quad \text{for every } w \text{ with } P(W = w) > 0$$

- **Structural violation:** a subgroup never receives one level — e.g. every long-term-care resident is vaccinated by policy. Restrict the population or change the question.
- **Random violation:** a sparse stratum is empty in one arm by chance.

Checkable from the data

Propensity-score overlap diagnostics make positivity visible — the TMLE walkthrough (Module G) inspects overlap directly.

Consistency — and What the Three Buy Us

A well-defined intervention: if $A = a$, the observed outcome *is* the counterfactual one, $Y = Y^{(a)}$ — “vaccinated” must mean one thing (which product, when, what schedule). The target-trial protocol makes this concrete.

Together they identify the effect from observable data

$$E[Y^{(a)}] = \sum_w E[Y \mid A = a, W = w] P(W = w)$$

Assumption	Checkable?	Defended by
Exchangeability	no	DAG + sensitivity analysis
Positivity	yes	overlap diagnostics
Consistency	no	target-trial protocol

Why it matters here: no estimator — however modern — can rescue a failed assumption.

We do not solve this with a better model first.

We solve it by designing the question.

Poll Bank

Vote in chat: A, B, or C

Poll 1 – Classify the Variable

In a study of a new anticoagulant versus an older one, **baseline kidney function** predicts both which drug the physician prescribes and the risk of a bleeding event.

Baseline kidney function is a:

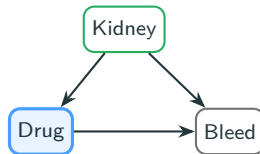
- A. Confounder
- B. Mediator
- C. Collider

Poll 1 – Solution

A – Confounder.

Baseline kidney function is measured *before* treatment and causes both the drug choice and bleeding – a common cause (fork).

Action: adjust to close the backdoor.



green = adjust for it

Poll 2 – Classify the Variable

A blood-pressure drug reduces stroke risk. You have a measurement of **on-treatment blood pressure at 6 months**. You want the *total* effect of the drug on stroke.

On-treatment blood pressure is a:

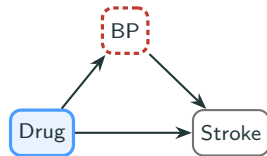
- A. Confounder – adjust for it
- B. Mediator – do not adjust
- C. Collider – do not adjust

Poll 2 – Solution

B – Mediator.

On-treatment blood pressure lies on the path (drug $\rightarrow$ BP $\rightarrow$ stroke) and is measured *after* time zero.

Do not adjust for a total effect.



red box = do not condition

Poll 3 – Classify the Variable

Among **hospitalized** patients only, a drug's side effects and the underlying disease both raise the chance of admission. You analyze within the hospitalized group.

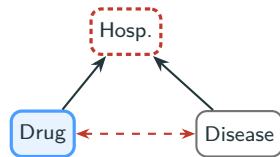
Hospitalization (the variable you restricted on) is a:

- A. Confounder
- B. Mediator
- C. Collider

C – Collider.

Side effects and disease both cause admission, so both arrows point *into* hospitalization.

Restricting to it opens a fake link.



dashed = induced association

Poll 4 – The Tricky One

A variable Z is measured *before* treatment. It is caused by two unmeasured factors: one that also causes treatment, and one that also causes the outcome. Z itself does *not* cause A or Y .

Should you adjust for Z ?

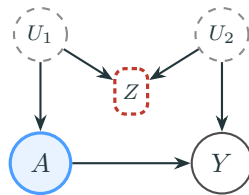
- A. Yes – it is pre-treatment, so it is safe
- B. No – adjusting opens a biasing path
- C. It never matters either way

Poll 4 – Solution

B – do not adjust (M-bias).

Z is a collider between two unmeasured causes. Conditioning on it opens the $U_1 \cdots U_2$ path and biases $A \rightarrow Y$.

“Pre-treatment” $\neq$ “safe.”



dashed circle = unmeasured

Poll 5 – What Is Wrong With This Comparison?

Patients who filled a prescription for the drug *within the first year* are labelled “treated”; everyone else is “untreated.” Survival is compared from cohort entry.

The main problem is:

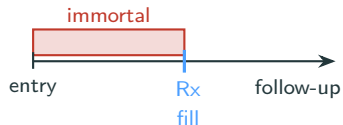
- A. Too small a sample
- B. Immortal time – the treated had to survive long enough to fill it
- C. Measurement error in the outcome

Poll 5 – Solution

B – Immortal time.

To be labelled “treated,” a patient must survive to the fill date. That guaranteed window biases toward the drug.

Fix: start time zero at treatment assignment.



survival in the shaded window is guaranteed

Poll 6 – Is This Causal?

A study finds that older adults who received the **pneumonia vaccine** have *higher* rates of pneumonia than those who did not.

The most likely explanation is:

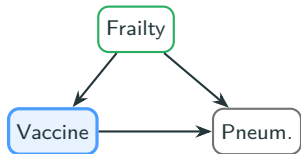
- A. The vaccine causes pneumonia
- B. Sicker, higher-risk patients were the ones vaccinated
- C. Random chance

Poll 6 – Solution

B – confounding by indication.

Frailer patients are targeted for the vaccine, so frailty raises both vaccination and pneumonia risk.

Tell: the gap predates vaccination.



frailty confounds the crude contrast

Poll 7 – Which Is Safe to Adjust For?

In a statin study you could adjust for LDL cholesterol measured *at baseline*, or LDL measured *at 6 months on treatment*. You want the **total** effect of the statin on heart attacks.

Which is safe to adjust for?

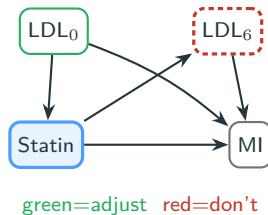
- A. Baseline LDL
- B. On-treatment LDL
- C. Both, to be safe

Poll 7 – Solution

A – baseline LDL only.

Baseline LDL is a pre-treatment confounder (adjust). On-treatment LDL is a mediator (do not).

Same variable, two roles – set by timing.



Poll 8 – The Paradox

Pooled together, patients on the new drug have *higher* mortality than those on the old drug. But within *every* disease-severity stratum, the new drug has *lower* mortality.

Which comparison reflects the drug's effect?

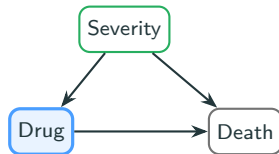
- A. The pooled comparison
- B. The within-severity comparison
- C. Neither can be trusted

Poll 8 – Solution

B – within-severity, here.

Severity is a pre-treatment confounder: sicker patients got the new drug. Stratifying removes the bias.

Catch: only because the stratifier is a confounder.



stratify on a confounder, not a mediator

Poll 9 – The Placebo Patients

In a randomized trial, patients who took their **placebo** as prescribed had substantially *lower* mortality than patients who did not adhere to the placebo.

The best explanation is:

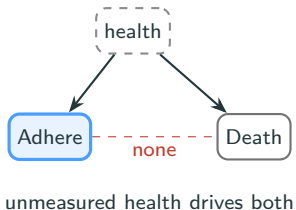
- A. The placebo had a real effect
- B. Adherence is a marker of healthier behaviour overall
- C. The randomization failed

Poll 9 – Solution

B – healthy-adherer bias.

Adhering even to placebo marks healthier, more health-conscious people. Adherence is a confounded proxy, not a cause.

Randomization protects the *assigned* contrast only.



Poll 10 – The Painkiller and the Cancer

A painkiller is prescribed for early back pain. Months later, some of those patients are diagnosed with spinal cancer – which had been causing the pain all along. In the data, the painkiller is associated with cancer.

This is:

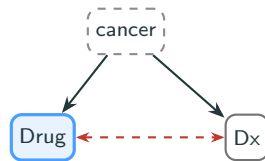
- A. A real carcinogenic effect
- B. Reverse causation: early symptoms of the outcome triggered the drug
- C. Confounding by age

Poll 10 – Solution

B – reverse causation.

Undiagnosed cancer caused the pain that led to the drug. The arrow runs outcome $\rightarrow$ exposure.

Fix: a lag/exclusion window before time zero.



the “effect” is driven by latent disease

Poll 11 – The Table 2 Fallacy

A model estimates a drug's effect on stroke, adjusting for age, smoking, and hypertension. The results table also reports the **smoking** coefficient and the paper reads it as “the effect of smoking on stroke.”

Is that smoking coefficient a valid causal effect of smoking?

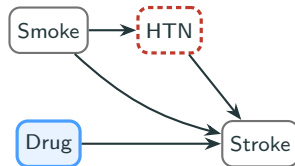
- A. Yes – it was adjusted for the same covariates
- B. Not generally – the adjustment set was chosen for the drug, not for smoking
- C. Yes, always

Poll 11 – Solution

B – not generally valid.

The set that identifies the *drug* effect need not identify smoking's: adjusting HTN (a mediator of smoking) blocks part of its own effect.

One model, one estimand.



adjusting HTN: right for Drug, wrong for Smoke

Poll 12 – The Feedback Loop

In an HIV study, CD4 count is *affected by* prior antiretroviral treatment, *predicts* whether treatment continues, and *predicts* mortality. You want the effect of sustained treatment on mortality.

Standard regression adjusting for CD4:

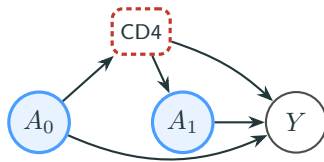
- A. Works fine – CD4 is a confounder, so adjust for it
- B. Is biased – CD4 both confounds later treatment *and* is affected by earlier treatment
- C. Should ignore CD4 entirely

Poll 12 – Solution

B – time-varying confounder.

CD4 is affected by early treatment *and* confounds later treatment. Adjust and you block/collide; omit and you confound.

Use g-methods.



CD4: mediator of A_0 , confounder of A_1

Poll 13 – Spot the Instrument

You want an **instrumental variable** for whether a patient receives the new drug. Which candidate is the most plausible?

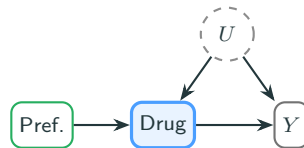
- A. Baseline disease severity
- B. The prescribing physician's general preference for the new drug
- C. The patient's on-treatment lab values

Poll 13 – Solution

B – physician preference.

An instrument moves treatment but reaches Y *only through* it. Severity is a confounder; labs are a mediator.

Assumptions are strong, untestable.



instrument sidesteps unmeasured U

Poll 14 – No Comparison Group

Clinical guidelines mandate the drug for *everyone* with severe disease, so in your data there are **zero untreated** severe patients.

For the severe subgroup, you:

- A. Can still estimate the effect by adjusting
- B. Cannot estimate it – there is no comparison group
- C. Should impute the missing untreated patients

Poll 14 – Solution

B – positivity violation.

Identification needs both arms within each stratum. With no untreated severe patients, that effect is not in the data.

Imputing the arm is extrapolation.

	Treated	Untreated
mild	• • •	• •
severe	• • •	(none)

no untreated severe patients = no contrast

Poll 15 – The Completers Analysis

In a cohort, treated patients who do poorly tend to drop out early, while untreated patients who do poorly stay in. You analyze only the patients who **completed** follow-up.

Restricting to completers:

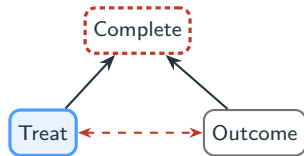
- A. Is fine – completers have the cleanest data
- B. Induces selection bias – completion is a collider
- C. Only reduces power

Poll 15 – Solution

B – collider selection bias.

Completing follow-up is a common effect of treatment and prognosis. Keeping only completers conditions on that collider.

Same structure as Poll 3.



conditioning on completion opens a path

Poll 16 – Screening “Adds a Year”

Patients whose cancer is found by **screening** survive on average 3 years from diagnosis; those found after **symptoms** survive 2 years. Screening appears to add a year of life.

The most likely explanation is:

- A. Screening genuinely extends life by a year
- B. Lead-time bias – earlier diagnosis just starts the clock sooner
- C. The screened cancers were less aggressive

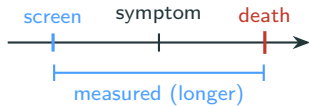
Poll 16 – Solution

B – lead-time bias.

Screening moves the diagnosis date earlier.

Survival *from diagnosis* stretches even if death never moves.

An artefact of where the clock starts.



earlier start, same death date

References

- [1] Hernán MA, Robins JM. Using big data to emulate a target trial when a randomized trial is not available. *American Journal of Epidemiology*, 183(8):758–764, 2016.
- [2] Hernán MA, Robins JM. *Causal Inference: What If*. Chapman & Hall/CRC, 2020.
- [3] Pearl J, Mackenzie D. *The Book of Why: The New Science of Cause and Effect*. Basic Books, 2018.
- [4] Bonten MJM, Huijts SM, Bolkenbaas M, et al. Polysaccharide conjugate vaccine against pneumococcal pneumonia in adults. *New England Journal of Medicine*, 372(12):1114–1125, 2015.